

The 3-leg copy tensor under the group Fourier transform

Q' (real space $\rightarrow$ irrep basis) turns it into the Clebsch–Gordan / fusion tensor

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1 Setup

Let G be a finite group (the space group on the BvK supercell) with unitary irreps $\{\rho\}$, $\dim \rho = d_\rho$, matrices $\rho(g)_{ab}$. Let X be the finite set of real-space grid points; G acts on X by permutation, $g \cdot x$. On $\mathbb{C}^X = \{\text{functions on } X\}$ this is the permutation representation

$$\pi(g) : (\pi(g)f)(x) = f(g^{-1}x).$$

Decompose $\mathbb{C}^X \cong \bigoplus_\rho V_\rho \otimes \mathbb{C}^{m_\rho}$ (m_ρ = multiplicity of ρ). The **symmetry-adapted (irrep) basis** $|\rho ac\rangle$ ($a = 1 \dots d_\rho$ partner, $c = 1 \dots m_\rho$ copy) is the one in which π is block-diagonal, and Q' is the unitary that produces it:

$$|\rho ac\rangle = \sum_{x \in X} u_a^{\rho c}(x) |x\rangle, \quad u_a^{\rho c}(x) = \langle x | \rho ac \rangle, \quad Q'_{x,(\rho ac)} = u_a^{\rho c}(x).$$

(For real-space input, $Q' = Q(F \otimes \mathbb{K})$: F is the cell DFT, Q the point-group reduction.)

The 3-leg copy tensor is the pointwise multiplication of $\mathbb{C}^X$ (the ISDF fusion vertex):

$$\delta_{xyz} = [x = y = z], \quad m^{\text{cp}}(|x\rangle \otimes |y\rangle) = \delta_{xy}|x\rangle.$$

The object to compute — the copy tensor with all three legs mapped to the irrep basis:

$$C_{\rho_1 a_1 c_1, \rho_2 a_2 c_2}^{\rho_3 a_3 c_3} := \langle \rho_3 a_3 c_3 | m^{\text{cp}}(|\rho_1 a_1 c_1\rangle \otimes |\rho_2 a_2 c_2\rangle) \rangle = \sum_{x \in X} \overline{u_{a_3}^{\rho_3 c_3}(x)} u_{a_1}^{\rho_1 c_1}(x) u_{a_2}^{\rho_2 c_2}(x). \quad (\star)$$

Theorem. C factorizes as a group-determined Clebsch–Gordan part on the partners *times* a reduced tensor on the copies:

$$C_{\rho_1 a_1 c_1, \rho_2 a_2 c_2}^{\rho_3 a_3 c_3} = \sum_{\alpha=1}^{N_{\rho_1 \rho_2}^{\rho_3}} F_{\rho_1 \rho_2}^{\rho_3 \alpha}(c_1, c_2, c_3) \langle \rho_3 a_3 ; \alpha | \rho_1 a_1 ; \rho_2 a_2 \rangle$$

where $\langle \cdot | \cdot \rangle$ are Clebsch–Gordan coefficients, $N_{\rho_1 \rho_2}^{\rho_3}$ the fusion multiplicity, and F depends only on (ρ_i, α, c_i) — not on the partners a_i . In particular $C = 0$ unless $\rho_3 \subseteq \rho_1 \otimes \rho_2$ (the fusion rule).

2 Proof, step by step

Step 1 (covariance of Q'). By definition of the symmetry-adapted basis, $\pi(g)|\rho ac\rangle = \sum_{a'} \rho(g)_{a'a} |\rho a' c\rangle$ — π rotates the partner a by the irrep matrix and is the identity on the copy c . (This is exactly $Q'^{\dagger} \pi(g) Q' = \bigoplus_\rho \rho(g) \otimes \mathbb{K}_{m_\rho}$.)

Step 2 (equivariance of the copy tensor). Pointwise multiplication commutes with the permutation action:

$$(\pi(g)f \cdot \pi(g)h)(x) = f(g^{-1}x)h(g^{-1}x) = (fh)(g^{-1}x) = (\pi(g)(fh))(x),$$

so $m^{\text{CP}} \circ (\pi(g) \otimes \pi(g)) = \pi(g) \circ m^{\text{CP}}$. Thus m^{CP} is a G -intertwiner $\mathbb{C}^X \otimes \mathbb{C}^X \rightarrow \mathbb{C}^X$. Equivalently $m^{\text{CP}} = \pi(g)^{-1} m^{\text{CP}} (\pi(g) \otimes \pi(g))$.

Step 3 (C is an invariant tensor). Insert Step 2 into $(\star)$ and use Step 1 on each ket/bra (with $\pi(g)^{-1} = \pi(g)^\dagger$):

$$C_{\rho_1 a_1 c_1, \rho_2 a_2 c_2}^{\rho_3 a_3 c_3} = \sum_{a'_1 a'_2 a'_3} \overline{\rho_3(g)_{a'_3 a_3}} \rho_1(g)_{a'_1 a_1} \rho_2(g)_{a'_2 a_2} C_{\rho_1 a'_1 c_1, \rho_2 a'_2 c_2}^{\rho_3 a'_3 c_3} \quad \forall g \in G.$$

Fix $(\rho_1, \rho_2, \rho_3, c_1, c_2, c_3)$ and read C as a linear map $B : V_{\rho_1} \otimes V_{\rho_2} \rightarrow V_{\rho_3}$, $B(e_{a_1} \otimes e_{a_2}) = \sum_{a_3} C_{\dots a_1 a_2}^{\dots a_3} e_{a_3}$. The identity above says exactly $B(\rho_1(g) \otimes \rho_2(g)) = \rho_3(g) B$, i.e.

$$B \in \text{Hom}_G(V_{\rho_1} \otimes V_{\rho_2}, V_{\rho_3}).$$

Step 4 (Schur on the tensor product). The space of such intertwiners has dimension equal to the multiplicity of ρ_3 in $\rho_1 \otimes \rho_2$,

$$\dim \text{Hom}_G(V_{\rho_1} \otimes V_{\rho_2}, V_{\rho_3}) = N_{\rho_1 \rho_2}^{\rho_3} = \frac{1}{|G|} \sum_g \chi_{\rho_1}(g) \chi_{\rho_2}(g) \overline{\chi_{\rho_3}(g)},$$

and a canonical basis of it is the set of **Clebsch–Gordan intertwiners** $\{B^\alpha\}_{\alpha=1}^{N_{\rho_1 \rho_2}^{\rho_3}}$, with matrix elements the CG coefficients $\langle \rho_3 a_3; \alpha | \rho_1 a_1; \rho_2 a_2 \rangle$. (If $N = 0$, the space is $\{0\}$: the fusion rule.)

Step 5 (expand C in that basis). Since $B \in \text{Hom}_G(\dots)$ it is a unique linear combination of the B^α ; the coefficients are numbers depending only on what was held fixed, namely $(\rho_1, \rho_2, \rho_3, \alpha)$ and the copies (c_1, c_2, c_3) :

$$C_{\rho_1 a_1 c_1, \rho_2 a_2 c_2}^{\rho_3 a_3 c_3} = \sum_\alpha F_{\rho_1 \rho_2}^{\rho_3 \alpha}(c_1, c_2, c_3) \langle \rho_3 a_3; \alpha | \rho_1 a_1; \rho_2 a_2 \rangle.$$

This is the boxed statement. Projecting on the CG basis (orthogonality $\sum_{a_1 a_2 a_3} \langle \rho_3 a_3; \alpha | \dots \rangle \overline{\langle \rho_3 a_3; \beta | \dots \rangle} = d_{\rho_3} \delta_{\alpha \beta}$) gives the reduced coefficient explicitly (Wigner–Eckart form):

$$F_{\rho_1 \rho_2}^{\rho_3 \alpha}(c_1, c_2, c_3) = \frac{1}{d_{\rho_3}} \sum_{a_1 a_2 a_3} \overline{\langle \rho_3 a_3; \alpha | \rho_1 a_1; \rho_2 a_2 \rangle} C_{\rho_1 a_1 c_1, \rho_2 a_2 c_2}^{\rho_3 a_3 c_3}. \quad \blacksquare$$

Reading. The partner dependence (a_1, a_2, a_3) is *entirely* the group's CG coefficients; the material/ X -dependent content sits in the reduced tensor F on the copies. The copy tensor, Fourier-transformed by Q' , is the fusion tensor of G .

3 Abelian special case (recovers modulo addition)

Let $G = \mathbb{Z}_N$ act on $X = \mathbb{Z}_N$ by translation (the pure translation / momentum case). All irreps are 1-dimensional characters $\chi_k(x) = e^{2\pi i k x / N}$, $k = 0, \dots, N-1$: no partner index, $d_\rho = 1$, and (regular

representation) each appears once, $m_\rho = 1$, so no copy index. The symmetry-adapted basis is the Fourier basis $u_k(x) = e^{2\pi i k x / N} / \sqrt{N}$. Then $(\star)$ is a direct sum:

$$C_{k_1 k_2}^{k_3} = \sum_{x=0}^{N-1} \overline{u_{k_3}(x)} u_{k_1}(x) u_{k_2}(x) = \frac{1}{N^{3/2}} \sum_x e^{2\pi i (k_1 + k_2 - k_3)x / N} = \frac{1}{\sqrt{N}} [k_1 + k_2 \equiv k_3 \pmod{N}].$$

So $C = \frac{1}{\sqrt{N}} \delta_{k_1 + k_2 \equiv k_3}$ — the **modulo-addition tensor**. This matches the theorem: 1-dim irreps $\Rightarrow N_{k_1 k_2}^{k_3} = [k_1 + k_2 \equiv k_3]$ (fusion = group addition on $\widehat{G}$), the CG coefficient is the scalar $\delta_{k_1 + k_2 \equiv k_3}$, and $F = 1/\sqrt{N}$ (trivial reduced tensor since $m \equiv 1$). For a general abelian G -set (copies present) the same computation gives $C_{k_1 c_1, k_2 c_2}^{k_3 c_3} = F(c_1, c_2, c_3) \delta_{k_1 + k_2 \equiv k_3}$: modulo addition on the momenta, reduced data on the copies.

4 Why this is the ISDF story

The ISDF fusion vertex (co-density = $X \cdot X$, a pointwise product) is m^{cp} . Sandwiched by the space-group Q' it becomes the fusion tensor: momentum conservation is the abelian (translation) factor $\delta_{k_1 + k_2 \equiv k_3}$; the point group contributes the genuine CG coefficients on the partners plus the reduced F on the copies. This is exactly why the fusion *densifies* in the irrep basis (it is no longer a sparse δ , but a CG tensor), and why full space-group symmetry gives momentum- $\delta \times$ point-group-CG $\times$ reduced-tensor.